

MD SEFAT

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RESEARCH INTERESTS

Machine learning and explainable AI applications in public health, with a focus on disease prediction, model interpretability, and health data analytics that supports clinical and epidemiological decision-making.

EDUCATION

Master of Data Science — Macquarie University, Sydney, Australia
28 July 2025 – Present

BSc in Computer Science and Engineering (CSE) — American International University–Bangladesh (AIUB)
Completed

RESEARCH EXPERIENCE

Machine Learning & Explainable AI for Diabetes Prediction — Research Project, Macquarie University
2026 – Present

- Reproduced and extended a published machine-learning study on diabetes prediction using healthcare datasets.
- Evaluated classification and ensemble methods — including XGBoost, Random Forest, SVM, Logistic Regression, and Bagging — applying SMOTE and ADASYN to address class imbalance.
- Investigated model interpretability using SHAP and LIME, assessing performance via accuracy, precision, recall, F1-score, and AUC.
- Reproduced key experimental results, including an XGBoost AUC of approximately 0.84, consistent with the original study.
- Extended the study to evaluate model performance and generalisation on additional diabetes-related datasets.

TECHNICAL SKILLS

Programming Languages: C#, Java, PHP, JavaScript, Python, R, SQL

Machine Learning & Data Analysis: Pandas, NumPy, scikit-learn, SHAP, LIME, XGBoost

Web & Software Development: HTML, CSS, .NET, PHP Frameworks

Tools & Platforms: Jupyter Notebook, Git, Excel

LANGUAGES

English (Fluent) • Bangla (Native)

REFERENCES

Available upon request.